

COORDINATE GEOMETRY

RECTANGULAR COORDINATE AXES

Let XOX' be a horizontal straight line and YOY' be a vertical straight line drawn through a point O in the plane of the paper. Then

the line XOX' is called x -axis

the line YOY' is called y -axis

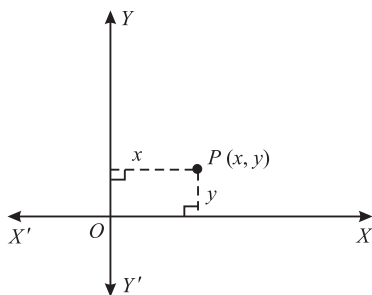
plane of paper is called xy -plane or cartesian plane.

x -axis and y -axis together are called co-ordinate axes or axis of reference.

The point O is called the origin.

Cartesian Coordinates

Position of any point in a cartesian plane can be described by their cartesian coordinates. The ordered pair of perpendicular distances first from y -axis and second from x -axis of a point P is called cartesian coordinates of P .



If the cartesian coordinates of point P are (x, y) , then x is called abscissa or x -coordinate of P and y is called the ordinate or y -coordinate of point P .

SIGN CONVENTIONS IN THE xy -PLANE

- All the distances are measured from origin (O).
- All the distances measured along or parallel to x -axis but right side of origin are taken as $+ve$.
- All the distances measured along or parallel to x -axis but left side of origin are taken as $-ve$.
- All the distances measured along or parallel to y -axis but above the origin are taken as $+ve$.
- All the distances measured along or parallel to y -axis but below the origin are taken as $-ve$.

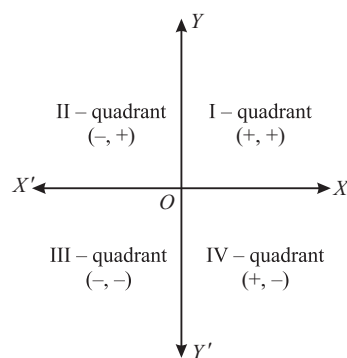
According to the Above Sign Conventions

- Coordinate of origin is $(0, 0)$
- Coordinate of any point on the x -axis but right side of origin is of the form $(x, 0)$, where $x > 0$.
- Coordinate of any point on the x -axis but left side of origin is of the form $(-x, 0)$, where $x > 0$.
- Coordinate of any point on the y -axis but above the origin is of the form $(0, y)$, where $y > 0$.
- Coordinate of any point on the y -axis but below the origin is of the form $(0, -y)$, where $y > 0$.

QUADRANTS OF xy -PLANE AND SIGN OF x AND y -COORDINATE OF A POINT IN DIFFERENT QUADRANTS

x and y -axis divide the xy -plane in four parts. Each part is called a quadrant.

The four quadrants are written as I-quadrant (XOY), II-quadrant (YOX'), III-quadrant ($X'OY'$) and IV-quadrant ($Y'OX$). Each of these quadrants shows the specific quadrant of the xy -plane as shown below:



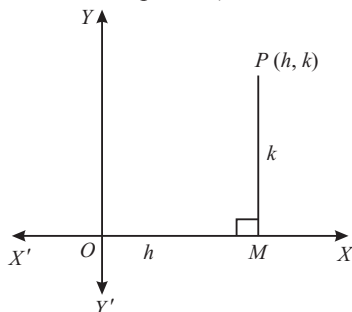
- Any of the four quadrants does not include any part of x or y -axis.
- In the first quadrant both x and y -coordinates of any point are $+ve$.
- In second quadrant x -coordinate of any point is $-ve$ but y -coordinate of any point is $+ve$.

- (iv) In third quadrant, both x and y -coordinates of any point are $-ve$.
- (v) In fourth quadrant, x -coordinate of any point is $+ve$ but y -coordinate of any point is $-ve$ as shown in the above diagram.

PLOTTING A POINT WHOSE COORDINATES ARE KNOWN

The point can be plotted by measuring its proper distances from both the axes. Thus, any point P whose coordinates are (h, k) can be plotted as follows:

- (i) Measure OM equal to h (i.e. x -coordinate of point P) along the x -axis.
- (ii) Now perpendicular to OM equal to k .
Mark point P above M such that PM is parallel to y -axis and $PM = k$ (i.e. y -coordinate of point P)



In this chapter, now we shall study to find the distance between two given points, section formula, mid-point formula, slope of a line, angles between two straight lines and equation of a line in different forms etc.

DISTANCE FORMULA

The distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by

$$PQ = \left| \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \right| \quad \text{or} \quad \left| \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \right|$$

$$\text{Distance of point } P(x, y) \text{ from the origin} = \left| \sqrt{x^2 + y^2} \right|$$

Illustration 1: If distance between the point $(x, 2)$ and $(3, 4)$ is 2, then find the value of x .

Solution:

$$2 = \left| \sqrt{(x-3)^2 + (2-4)^2} \right| \Rightarrow 2 = \left| \sqrt{(x-3)^2 + 4} \right|$$

Squaring both sides

$$4 = (x-3)^2 + 4 \Rightarrow x-3 = 0 \Rightarrow x = 3$$

Illustration 2: Find the distance between each of the following points :

$$A(-6, -1) \text{ and } B(-6, 11)$$

Solution: Here the points are $A(-6, -1)$ and $B(-6, 11)$

By using distance formula, we have

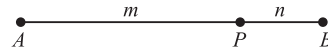
$$AB = \sqrt{\{-6 - (-6)\}^2 + \{11 - (-1)\}^2} = \sqrt{0^2 + 12^2} = 12$$

Hence, $AB = 12$ units.

SECTION FORMULA

Co-ordinates of a point which divides the line segment joining two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ in the ratio $m_1 : m_2$ are :

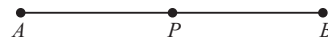
- (i) $\left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$, for internal division.



P divides AB internally in the ratio $m : n$

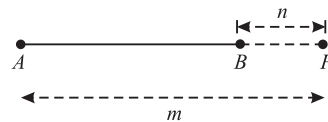
If $m_1 = m_2$, then the point P will be the mid point of PQ

$$\text{whose co-ordinates} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$



P is the mid-point of AB

- (ii) $\left(\frac{m_1 x_2 - m_2 x_1}{m_1 - m_2}, \frac{m_1 y_2 - m_2 y_1}{m_1 - m_2} \right)$, for external division



P divides AB externally in the ratio $m : n$

- (iii) When we need to find the ratio in which a point on a line segment divides it, we suppose the required ratio as $k : 1$ or $m/n : 1$.

Note:

- (i) Co-ordinates of any point on the line segment joining two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ are

$$\left(\frac{x_1 + \lambda x_2}{1 + \lambda}, \frac{y_1 + \lambda y_2}{1 + \lambda} \right), (\lambda \neq -1)$$

- (ii) **Division by axes:** Line segment joining the points (x_1, y_1) and (x_2, y_2) is divided by

(a) x -axis in the ratio $-y_1 / y_2$

(b) y -axis in the ratio $-x / x_2$

If ratio is positive division internally and if ratio is negative division is externally.

- (iii) **Division by a line:** Line $ax + by + c = 0$ divides the line joining the points (x_1, y_1) and (x_2, y_2) in the ratio

$$\left(-\frac{ax_1 + by_1 + c}{ax_2 + by_2 + c} \right).$$

Illustration 3: Find the ratio in which the line $3x + 4y = 7$ divides the line segment joining the points $(1, 2)$ and $(-2, 1)$.

$$\text{Solution: Ratio} = -\frac{3(1) + 4(2) - 7}{3(-2) + 4(1) - 7} = -\frac{4}{-9} = \frac{4}{9} = 4 : 9.$$

Illustration 4: Find the points of trisection of line joining the points $A(2, 1)$ and $B(5, 3)$.

$$\text{Solution: } (2, 1) \xrightarrow{\quad 1 \quad} \xrightarrow{\quad 2 \quad} (5, 3)$$

$A \quad P_1 \quad P_2 \quad B$
 $\xleftarrow{\quad 2 \quad} \xleftarrow{\quad 1 \quad}$

$$P_1(x, y) = \left(\frac{1 \times 5 + 2 \times 2}{1 + 2}, \frac{1 \times 3 + 2 \times 1}{1 + 2} \right) = \left(3, \frac{5}{3} \right)$$

$$P_2(x, y) = \left(\frac{2 \times 5 + 1 \times 2}{2 + 1}, \frac{2 \times 3 + 1 \times 1}{2 + 1} \right) = \left(4, \frac{7}{3} \right).$$

Illustration 5: Prove that points $A(1, 1)$, $B(-2, 7)$ and $C(3, -3)$ are collinear.

Solution: $AB = \sqrt{(1+2)^2 + (1-7)^2} = \sqrt{9+36} = 3\sqrt{5}$

$$BC = \sqrt{(-2-3)^2 + (7+3)^2} = \sqrt{25+100} = 5\sqrt{5}$$

$$CA = \sqrt{(3-1)^2 + (-3-1)^2} = \sqrt{4+16} = 2\sqrt{5}$$

Clearly, $BC = AB + AC$. Hence A, B, C are collinear.

Illustration 6: Find the ratio in which the join of $(-4, 3)$ and $(5, -2)$ is divided by (i) x -axis (ii) y -axis.

Solution:

(i) x -axis divides the join of (x_1, y_1) and (x_2, y_2) in the ratio of $-y_1 : y_2 = -3 : -2 = 3 : 2$.

(ii) y -axis divides, in the ratio of $-x_1 : x_2 \Rightarrow 4 : 5$.

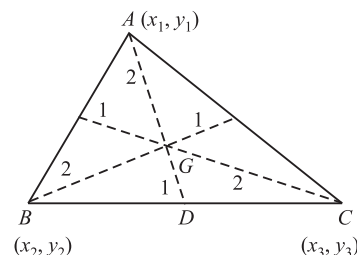
COORDINATES OF SOME PARTICULAR POINTS

Let $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$ are vertices of any triangle ABC , then

Centroid

Centroid is the point of intersection of the medians of a triangle. Centroid divides each median in the ratio of $2 : 1$.

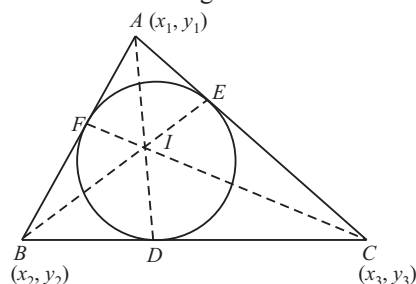
A median is a line segment joining the mid point of a side to its opposite vertex of a triangle.



Co-ordinates of centroid, $G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$

Incentre

Incentre is the point of intersection of internal bisectors of the angles of a triangle. Also incentre is the centre of the circle touching all the sides of a triangle.



Co-ordinates of incentre,

$$I = \left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c} \right),$$

where a, b, c are length of the sides opposite to vertices A, B, C respectively of triangle ABC .

(i) Angle bisector divides the opposite sides in the ratio of the sides included in the angle. For example

$$\frac{BD}{DC} = \frac{AB}{AC} = \frac{c}{b}.$$

(ii) Incentre divides the angle bisectors AD, BE and CF in the ratio $(b + c) : a$, $(c + a) : b$ and $(a + b) : c$ respectively.



Practice Exercise



Level - I

- If distance between the point $(x, 2)$ and $(3, 4)$ is 2, then the value of $x =$
 (a) 0 (b) 2
 (c) 3 (d) 4
- Find the mid-point of the line-segment joining two points $(3, 4)$ and $(5, 12)$.
 (a) $(-4, 8)$ (b) $(0, 8)$
 (c) $(4, 8)$ (d) $(4, 0)$
- The mid-point of the line segment joining the points $(-2, 4)$ and $(6, 10)$ is
 (a) $(2, 5)$ (b) $(2, 7)$
 (c) $(3, 7)$ (d) $(3, 8)$
- The points $A(-4, -1)$, $B(-2, -4)$, $C(4, 0)$ and $D(2, 3)$ are the vertices of a
 (a) Parallelogram (b) Rectangle
 (c) Rhombus (d) Square
- The line $x + y = 4$ divides the line joining the points $(-1, 1)$ and $(5, 7)$ in the ratio
 (a) $2 : 1$ (b) $1 : 2$
 (c) $1 : 2$ externally (d) None of these
- If $A(3, 5)$, $B(-3, -4)$, $C(7, 10)$ are the vertices of a parallelogram taken in the order, then the co-ordinates of the fourth vertex are
 (a) $(10, 19)$ (b) $(15, 10)$
 (c) $(19, 10)$ (d) $(15, 19)$
- The centroid of a triangle, whose vertices are $(2, 1)$, $(5, 2)$ and $(3, 4)$ is
 (a) $\left(\frac{8}{3}, \frac{7}{3}\right)$ (b) $\left(\frac{10}{3}, \frac{7}{3}\right)$
 (c) $\left(-\frac{10}{3}, \frac{7}{3}\right)$ (d) $\left(\frac{10}{3}, -\frac{7}{3}\right)$
- The incentre of the triangle with vertices $(1, \sqrt{3})$, $(0, 0)$ and $(2, 0)$ is
 (a) $\left(1, \frac{\sqrt{3}}{2}\right)$ (b) $\left(\frac{2}{3}, \frac{1}{\sqrt{3}}\right)$
 (c) $\left(\frac{2}{3}, \frac{\sqrt{3}}{2}\right)$ (d) $\left(1, \frac{1}{\sqrt{3}}\right)$
- The centroid of the triangle whose vertices are $(3, 10)$, $(7, 7)$, $(-2, 1)$ is
 (a) $(8/3, 6)$ (b) $(6, 8/3)$
 (c) $(-4, -7/3)$ (d) None of these
- The coordinates of the centroid G of a triangle with vertices at $(3, 7)$, $(5, 5)$ and $(-3, 2)$ is
 (a) $(10/3, 14/3)$ (b) $(10/3, 10/3)$
 (c) $(5/3, 14/3)$ (d) $(11/3, 10/3)$
- The coordinates of a point which divides the join of $(5, -5)$ and $(2, -3)$ in the ratio $4 : 3$, externally, are:
 (a) $(3, 4)$ (b) $(-7, 3)$
 (c) $(-7, 9)$ (d) $(8, 3)$
- Distance between $P(x, y)$ and $Q(3, -6)$ is 10 units and x is positive integer, then $x =$
 (a) 3 (b) 9
 (c) 7 (d) 11
- The vertices of a parallelogram in order are $A(1, 2)$, $B(4, y)$, $C(x, 6)$, $D(3, 6)$, then $(x, y) =$
 (a) $(6, 3)$ (b) $(3, 6)$
 (c) $(5, 6)$ (d) $(1, 4)$
- The point which divides the line segment joining the points $(7, -6)$ and $(3, 4)$ in ratio $1 : 2$ internally lies in the
 (a) I quadrant (b) II quadrant
 (c) III quadrant (d) IV quadrant
- How many squares are possible if two of the vertices of a quadrilateral are $(1, 0)$ and $(2, 0)$?
 (a) 1 (b) 2
 (c) 3 (d) 4
- In what ratio is the line segment made by the points $(7, 3)$ and $(-4, 5)$ divided by the y -axis?
 (a) $2 : 3$ (b) $4 : 7$
 (c) $3 : 5$ (d) $7 : 4$
- If the coordinates of the mid-point of the line segment joining the points $(2, 1)$ and $(1, -3)$ is (x, y) , then the relation between x and y can be best described by
 (a) $3x + 2y = 5$ (b) $6x + y = 8$
 (c) $5x - 2y = 4$ (d) $2x - 5y = 4$
- Points $(4, -1)$, $(6, 0)$, $(7, 2)$ and $(5, 1)$ are joined to be a vertex of a quadrilateral. What will be the structure?
 (a) Rhombus (b) Parallelogram
 (c) Square (d) Rectangle
- Find the third vertex of the triangle whose two vertices are $(-3, 1)$ and $(0, -2)$ and the centroid is the origin.
 (a) $(2, 3)$ (b) $\left(-\frac{4}{3}, \frac{14}{3}\right)$
 (c) $(3, 1)$ (d) $(6, 4)$
- If the origin gets shifted to $(2, 2)$, then what will be the new coordinates of the point $(4, -2)$?
 (a) $(-2, 4)$ (b) $(2, 4)$
 (c) $(4, 2)$ (d) $(2, -4)$

21. If the point $R(1, -2)$ divides externally the line segment joining $P(2, 5)$ and Q in the ratio $3 : 4$, what will be the coordinates of Q ?
 (a) $(-3, 6)$ (b) $(2, -4)$
 (c) $(3, 6)$ (d) $(1, 2)$
22. C is the mid-point of PQ , if P is $(4, x)$, C is $(y, -1)$ and Q is $(-2, 4)$, then x and y respectively are
 (a) -6 and 1 (b) -6 and 2
 (c) 6 and -1 (d) 6 and -2
23. A quadrilateral has the vertices at the points $(-4, 2)$, $(2, 6)$, $(8, 5)$ and $(9, -7)$. Show that the mid-points of the sides of this quadrilateral are the vertices of a parallelogram.
 (a) Rectangle (b) Square
 (c) Parallelogram (d) Rhombus
24. Find the ratio in which the point $(2, y)$ divides the join of $(-4, 3)$ and $(6, 3)$ and hence find the value of y
 (a) $2 : 3, y = 3$ (b) $3 : 2, y = 4$
 (c) $3 : 2, y = 3$ (d) $3 : 2, y = 2$
25. If $P\left(\frac{a}{3}, 4\right)$ is the mid-point of the line segment joining the points $Q(-6, 5)$ and $R(-2, 3)$, then the value of a is
 (a) -4 (b) -12
 (c) 12 (d) -6
26. The ratio in which the line $2x + y - 4 = 0$ divides the line segment joining the points $A(2, -2)$ and $B(3, 7)$ is
 (a) $3 : 7$ (b) $4 : 7$
 (c) $2 : 9$ (d) $4 : 9$
27. Which of the following points is the nearest to the origin?
 (a) $(0, -6)$ (b) $(-8, 0)$
 (c) $(-3, -4)$ (d) $(7, 0)$
28. If the points $(1, 1)$, $(-1, -1)$ and $(-\sqrt{3}, k)$ are vertices of an equilateral triangle then the value of k will be :
 (a) 1 (b) -1
 (c) $\sqrt{3}$ (d) $-\sqrt{3}$
29. The points $(3, 0)$, $(-3, 0)$, $(0, -3\sqrt{3})$ are the vertices of
 (a) equilateral triangle (b) isosceles triangle
 (c) right triangle (d) scalene triangle
30. Ratio in which the line $3x + 4y = 7$ divides the line segment joining the points $(1, 2)$ and $(-2, 1)$ is
 (a) $3 : 5$ (b) $4 : 6$
 (c) $4 : 9$ (d) None of these
31. If the area of a triangle with vertices $(-3, 0)$, $(3, 0)$ and $(0, k)$ is 9 sq unit, then what is the value of k ?
 (a) 3 (b) 6
 (c) 9 (d) 12
32. The line $y = 0$ divides the line joining the points $(3, -5)$ and $(-4, 7)$ in the ratio
 (a) $3 : 4$ (b) $4 : 5$
 (c) $5 : 7$ (d) $7 : 9$
33. The line passing through the points $(-2, 8)$ and $(5, 7)$
 [SSC-Sub. Ins.-2012]
 (a) does not cut any axes (b) cuts x-axis only
 (c) cuts y-axis only (d) cuts both the axes

Level - II

1. The fourth vertex of a rectangle whose other vertices are $(4, 1)$, $(7, 4)$ and $(13, -2)$ is
 (a) $(10, -5)$ (b) $(10, 5)$
 (c) $(-10, 5)$ (d) $(-10, -5)$
2. The coordinates of vertices A and B of an equilateral triangle ABC are $(-4, 0)$ and $(4, 0)$ respectively. Which of the following could be coordinates of C
 (a) $(0, 2\sqrt{3})$ (b) $(0, 4)$
 (c) $(0, 4\sqrt{3})$ (d) $(0, 3)$
3. The three vertices of a parallelogram are $A(3, -4)$, $B(-2, 1)$ and $C(-6, 5)$. Which of the following cannot be the fourth one
 (a) $(-1, 0)$ (b) $(7, -8)$
 (c) $(1, -5)$ (d) All of these
4. The mid-points of sides of a triangle are $(2, 1)$, $(-1, -3)$ and $(4, 5)$. Then the coordinates of its vertices are:
 (a) $(7, 9)$, $(-3, -7)$, $(1, 1)$ (b) $(-3, -7)$, $(1, 1)$, $(2, 3)$
 (c) $(1, 1)$, $(2, 3)$, $(-5, 8)$ (d) None of these
5. The point whose abscissa is equal to its ordinate and which is equidistant from the points $(1, 0)$ and $(0, 3)$ is
 (a) $(1, 1)$ (b) $(2, 2)$
 (c) $(3, 3)$ (d) $(4, 4)$
6. If the point dividing internally the line segment joining the points (a, b) and $(5, 7)$ in the ratio $2 : 1$ be $(4, 6)$, then
 (a) $a = 1, b = 2$ (b) $a = 2, b = -4$
 (c) $a = 2, b = 4$ (d) $a = -2, b = 4$
7. The distance of point of intersection of $2X - 3Y + 13 = 0$ and $3X + 7Y - 15 = 0$ from $(4, -5)$, will be
 (a) 10 units (b) 12 units
 (c) 11 units (d) None of these
8. $A(-2, 4)$ and $B(-5, -3)$ are two points. The coordinates of a point P on Y axis such that $PA = PB$, are
 (a) $(3, 4)$ (b) $(0, 9)$
 (c) $(9, 0)$ (d) $(0, -1)$
9. The centroid of a triangle formed by $(7, p)$, $(q, -6)$, $(9, 10)$ is $(6, 3)$. Then $p + q$
 (a) 6 (b) 5
 (c) 7 (d) 8

10. If the three vertices of a rectangle taken in order are the points $(2, -2)$, $(8, 4)$ and $(5, 7)$. The coordinates of the fourth vertex is
 (a) $(1, 1)$ (b) $(1, -1)$
 (c) $(-1, 1)$ (d) None of these
11. If $P(1, 2)$, $Q(4, 6)$, $R(5, 7)$ and $S(a, b)$ are the vertices of a parallelogram $PQRS$, then
 (a) $a = 2, b = 4$ (b) $a = 3, b = 4$
 (c) $a = 2, b = 3$ (d) $a = 3, b = 5$
12. Find the coordinates of the points that trisect the line segment joining $(1, -2)$ and $(-3, 4)$
 (a) $\left(-\frac{1}{3}, 0\right)$ (b) $\left(-\frac{5}{3}, 2\right)$
 (c) Both (a) and (b) (d) None of these
13. If the mid-point of the line joining $(3, 4)$ and $(p, 7)$ is (x, y) and $2x + 2y + 1 = 0$, then what will be the value of p ?
 (a) 15 (b) $-\frac{17}{2}$
 (c) -15 (d) $\frac{17}{2}$
14. Two vertices of a triangle are $(5, -1)$ and $(-2, 3)$. If the orthocentre of the triangle is the origin, what will be the coordinates of the third point?
 (a) $(4, 7)$ (b) $(-4, 7)$
 (c) $(-4, -7)$ (d) $(4, -7)$
15. A point P is equidistant from $A(3, 1)$ and $B(5, 3)$ and its abscissa is twice its ordinate, then its co-ordinates are.
 (a) $(2, 1)$ (b) $(1, 2)$
 (c) $(4, 2)$ (d) $(2, 4)$
16. If $(-1, -1)$ and $(3, -1)$ are two opposite corners of a square, the other two corners are
 (a) $(2, 0), (-2, 2)$ (b) $(2, -2), (0, 2)$
 (c) $(3, 0), (4, -2)$ (d) None of these
17. What is the perimeter of the triangle with vertices $A(-4, 2)$, $B(0, -1)$ and $C(3, 3)$?
 (a) $7 + 3\sqrt{2}$ (b) $10 + 5\sqrt{2}$
 (c) $11 + 6\sqrt{2}$ (d) $5 + \sqrt{2}$
18. The area (in sq. unit) of the triangle formed by the three graphs of the equations $x = 4$, $y = 3$, and $3x + 4y = 12$, is
 [SSC CGL-2012]
 (a) 12 (b) 10
 (c) 6 (d) 8
19. The radius of the circumcircle of the triangle made by x -axis, y -axis and $4x + 3y = 12$ is
 [SSC CGL-2012]
 (a) 2 unit (b) 2.5 unit
 (c) 3 unit (d) 4 unit
20. The total area (in sq. unit) of the triangles formed by the graph of $4x + 5y = 40$, x -axis, y -axis and $x = 5$ and $y = 4$ is
 [SSC CGL-2014]
 (a) 10 (b) 20
 (c) 30 (d) 40

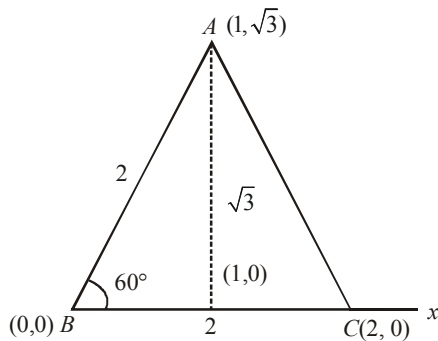


Hints & Solutions



Level-I

1. (c) $2 = \sqrt{(x-3)^2 + (2-4)^2} \Rightarrow 2 = \sqrt{(x-3)^2 + 4}$
Squaring both sides
 $4 = (x-3)^2 + 4 \Rightarrow x-3 = 0 \Rightarrow x = 3$
2. (c) Let $A(3, 4)$ and $B(5, 12)$ be the given points.
Let $C(x, y)$ be the mid-point of AB . Using mid-point formula, we have, $x = \frac{3+5}{2} = 4$ and $y = \frac{4+12}{2} = 8$
 $\therefore C(4, 8)$ are the co-ordinates of the mid-point of the line segment joining two points $(3, 4)$ and $(5, 12)$.
3. (b) 4. (b)
5. (b) Ratio $= -\left(\frac{-1+1-4}{5+7-4}\right) = \frac{1}{2}$
6. (d) Mid point of $A(3, 5)$ and $C(7, 10) = M\left(5, \frac{15}{2}\right)$
 $\therefore$ Mid points of $BD = M\left(5, \frac{15}{2}\right)$
 $B(-5, -4)$ and $D(x, y)$
 $\therefore \frac{-5+x}{2} = 5, x = 10+5 = 15$
 $\frac{-4+y}{2} = \frac{15}{2}, y = 15+4 = 19$
Co-ordinates of fourth vertex $D = (15, 19)$
7. (b) $x = \frac{2+5+3}{3} = \frac{10}{3}$ and $y = \frac{1+2+4}{3} = \frac{7}{3}$
8. (d) Clearly, the triangle is equilateral.



So, the incentre is the same as the centroid.

$$\therefore \text{Incentre} = \left(\frac{1+0+2}{3}, \frac{\sqrt{3}+0+0}{3}\right) = \left(1, \frac{1}{\sqrt{3}}\right)$$

9. (a) Centroid $= \left(\frac{3+7-2}{3}, \frac{10+7+1}{3}\right) = \left(\frac{8}{3}, 6\right)$
10. (c) Let G be (X, Y) , then $X = \{3+5+(-3)\}/3 = 5/3$
 $Y = (7+5+2)/3 = 14/3 \Rightarrow G$ is $(5/3, 14/3)$
11. (b) Let the ratio be $4:3$ or $4/3:1$.

$$\text{Now } X = \frac{\frac{4}{3} \times 2 - 5}{\frac{4}{3} - 1} = \frac{\frac{8}{3} - 5}{\frac{1}{3}} = \frac{-\frac{7}{3}}{\frac{1}{3}} = -7$$

$$Y = \frac{\frac{4}{3}x - 3 + 5}{\frac{4}{3} - 1} = \frac{1}{\frac{1}{3}} = 3. \text{ Hence } (-7, 3)$$

12. (b)
13. (a) Mid-point of AC is $\left(\frac{1+x}{2}, \frac{2+6}{2}\right)$ i.e., $\left(\frac{1+x}{2}, 4\right)$;
Mid-point of BD is $\left(\frac{4+3}{2}, \frac{y+5}{2}\right)$
Since for a || gm, diagonals bisect each other
 $\therefore \frac{1+x}{2} = \frac{7}{2}$ and $\frac{y+5}{2} = 4 \Rightarrow x = 6, y = 3$
14. (d) 15. (c) 16. (d) 17. (b) 18. (a)
19. (c) 20. (d) 21. (c) 22. (a) 23. (c)
24. (c) Let the required ratio be $k:1$
Then, $2 = \frac{6k-4 \times 1}{k+1} \Rightarrow k = \frac{3}{2}$
 $\therefore$ The required ratio is $\frac{3}{2} :: 1 \Rightarrow 3:2$

$$\text{Also, } y = \frac{3 \times 3 + 2 \times 3}{3+2}$$

25. (d) 26. (d) 27. (c)
28. (c) The equilateral Δ has its sides equal.
Hence the distance between the vertices should be equal.

$$a = \sqrt{2^2 + 2^2} = \sqrt{(\sqrt{3}+1)^2 + k(k-1)^2} \Rightarrow k = \sqrt{3}$$

29. (a) Find the three lengths separately

$$AB = 6, BC = \sqrt{3^2 + (3\sqrt{3})^2} = 6,$$

$$AC = \sqrt{3^2 + (3\sqrt{3})^2} = 6$$

Hence, the point are the vertices of equilateral triangle.

30. (c) $-\frac{3(1)+4(2)-7}{3(-2)+4(1)-7} = -\frac{4}{-9} = \frac{4}{9}$

31. (a) Let the vertices of the ΔABC be $A(-3, 0), B(3, 0)$ and $C(0, k)$.
Given, area is 9

$$\Rightarrow 9 = \frac{1}{2} \{-3(-k) + 1(3k)\}$$

$$\Rightarrow 18 = 3k + 3k$$

$$\Rightarrow k = \frac{18}{6} = 3$$

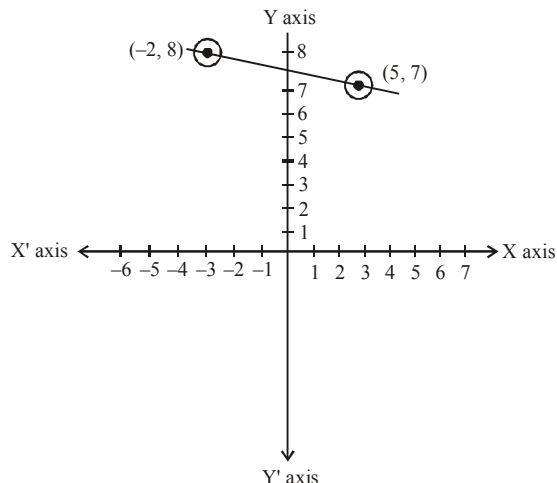
32. (c) Let $P(x, y)$ be the point of division that divides the line joining $(3, -5)$ and $(-4, 7)$ in the ratio of $k:1$

$$\text{Now, } y = \frac{7k-5}{k+1} \quad \dots (i)$$

Since, P lies on $y = 0$ or x -axis then, from eq. (i)

$$0 = \frac{7k-5}{k+1} \Rightarrow 7k = 5 \Rightarrow k = \frac{5}{7}$$

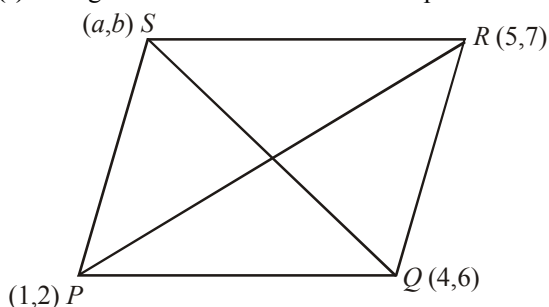
33. (c)



As indicated in the graph, the line passing through the points cuts Y-axis only.

Level-II

1. (a) 2. (c) 3. (d)
4. (a) $\frac{X_1 + X_2}{2} = 2$, $\frac{X_2 + X_3}{2} = -1$, $\frac{X_3 + X_1}{2} = 4$
 $\Rightarrow X_1 = 7, X_2 = -3, X_3 = 1$
 Similarly, y_1, y_2, y_3 can be found
5. (b) Let the point be (X, X) , so according to the condition $(X-1)^2 + (X-0)^2 = (X-0)^2 + (X-3)^2$
 $\Rightarrow 2X+1 = -6X+9 \Rightarrow X=2$
 Hence the point is $(2, 2)$
6. (c) $\frac{2 \times 5 + 1(a)}{2+1} = 4 \Rightarrow a=2$
 and $\frac{2 \times 7 + 1(b)}{2+1} = 6 \Rightarrow b=4$
7. (b) The point of intersection will be obtained by simultaneously solving the two equations and then by the distance formula, distance can be found.
8. (d) Take points P one by one and see which one $(0, -1)$ satisfies.
9. (c) By the given condition $\frac{7+q+9}{3} = 6$
 and $\frac{p-6+10}{3} = 3$
 $\Rightarrow q=2$ and $p=5 \therefore p+q=5+2=7$
10. (c) Let fourth vertex be (x, y) , then $\frac{x+8}{2} = \frac{2+5}{2}$
 and $\frac{y+4}{2} = \frac{-2+7}{2} \Rightarrow x=-1, y=1$
11. (c) Diagonals cut each other at middle points.



$$\text{Hence, } \frac{a+4}{2} = \frac{1+5}{2} \Rightarrow a=2$$

$$\frac{b+6}{2} = \frac{2+7}{2} \Rightarrow b=3$$

12. (c) 13. (c) 14. (c)
15. (c) Let the point be $P(2X, X)$. The choices we are left with are $(1, 2)$ and $(2, 4)$.

$$AP = \sqrt{(3-2X)^2 + (1-X)^2},$$

$$BP = \sqrt{(5-2X)^2 + (3-X)^2}$$

$$AP = BP. \text{ (only } (4, 2) \text{ satisfies)}$$

16. (d) We have the mid-point of diagonal $= (1, -1)$ which should be the mid point of the other two points as well and which is not satisfied by any given alternative.

17. (b) By using distance formula,

We have,

$$AB = \sqrt{(0+4)^2 + (-1-2)^2}$$

$$= \sqrt{16+9} = 5$$

$$BC = \sqrt{9+16} = 5$$

$$CA = \sqrt{49+(1)^2} = \sqrt{50}$$

$$= 5\sqrt{2}$$

$$\text{Hence, required perimeter} = AB + BC + CA$$

$$= 10 + 5\sqrt{2}$$

18. (c) $x=4$... (1)
 $y=3$... (2)
 $3x+4y=12$... (3)

Putting $x=0$ in 3rd equation we get $y=3$

Putting $y=0$ in 3rd equation we get $x=4$

The triangle will be formed by joining the points $(3, 0)$ and $(0, 4)$.

So, base = 3 and altitude = 4

$$\text{Area} = \frac{1}{2} \times b \times h \Rightarrow \frac{1}{2} \times 3 \times 4 = 6$$

19. (b) Putting $x=0$ in $4x+3y=12$ we get $y=4$
 Putting $y=0$ in $4x+3y=12$ we get $x=3$
 The triangle so formed is right angle triangle with points $(0, 0)$, $(4, 0)$, $(0, 3)$

So diameter is the hypotenus of triangle $= \sqrt{16+9}$
 $= 5$ unit

radius = 2.5 unit

20. (a)

Area of $\triangle ABC$

$$= \frac{1}{2} \times BC \times AC$$

$$= \frac{1}{2} \times (10-5) \times 4$$

$$= \frac{1}{2} \times 5 \times 4$$

$$\text{Area} = 10 \text{ sq unit.}$$

